过潜在的应用与现实世界相联系，因此在论文结尾处提及应用当然是合适的。然而，在讨论部分开始时陈述主要贡献或应用似乎是一股不断增长的趋势。在某些情况下，潜在应用本身就是研究的主要贡献或讨论部分的中心主题，这是在讨论开始时就陈述它们的一个很好的理由。经常查看你的目标期刊，看看它是如何应对当前的趋势和发展的。

如果我的工作没有任何明显的或当前的应用呢?

虽然很多研究没有明显的应用，但在放弃这个想法之前，最好先检查两个地方。首先，看看你的目标文章中的引言以及你自己的引言，因为在引言部分经常会提到可以以何种方式应用这种类型的研究。另一个可能的来源是你的目标文章的讨论或结论部分，因为这些地方可能会包括对未来潜在应用的推测。

内含的潜在应用表明你的研究价值超出了你的特定研究问题的目的。然而，研究有很多功能——它可能澄清一个理论，或者可能仅仅给一个发展中的领域添加新一层知识，所以你进行的这类研究可能在现阶段乃至永远都没有明确的应用。

句6、7和8："Some questions remain; for example, anecdotal evidence collected from patients during the study period suggested that there was a positive correlation between viral load at the acute stage and the severity of post-V2D health issues. However, this was not addressed formally during the study. Although many studies have

quantified viral load during acute V2D [17,22,27,33], a direct comparison with the post-viral period would provide valuable data to support such as correlation." **作者提到研究的局限性**(this was not addressed formally during the study)，**这引出对将来研究方向的建议**。

▶ 我如何决定在这里关注哪些局限性？我可以在讨论部分第一次提及研究的局限性吗？

大多数局限性来自研究期间遇到的问题，这些问题通常在相关小节中首次提及。然而，在这种情况下，anecdotal evidence（轶事证据）不是数据收集程序在形式上或意图上的一部分，也不是作为结果呈现的。这样的局限性与将来研究方向的建议相关，因此在讨论部分中第一次提及它是可以接受的。

另一个在讨论中首次提及局限性的可能原因是：你发现需要进一步工作才能得出结论。这是一个可以在讨论中提到的局限性，并与邀请研究界继续研究并取得进一步进展相关联。

▶ 为什么我要尝试确定未来工作的方向——为什么不让研究人员自己决定呢？

好的研究工作很少会本身自己结束，在大多数情况下，它们为研究工作开辟了方向。邀请科研团体以某种确定的方式关注你的工作有很多好处。首先，它为研究人员提供了一个合理的、明确的课题，这比一个模糊的建议更有吸引力，因此更有可能实施。其次，它促进了你的研究的直接连续性。跟随你的

研究工作的研究都会引用你的论文，这有利于增强你的研究地位。另外，一篇解决你的研究困难或局限性的论文有可能会为你目前和将来的工作提供有用的数据。因此，此时明显的问题是：研究工作的下一步应该往哪走？

句9："V2D RNA is known to persist in the bodies of survivors long after the disease onset[30,42], and the results of our study provide compelling evidence that follow-up of all convalescent patients is warranted for at least 10 months after discharge from hospital." **作者提及文献中的背景事实信息**（is known to persist）**以支持研究结论**。

▶ 即使在这个后期阶段，我还需要添加更多背景信息吗？

如上所述，留意目标文章讨论部分中每个引文的功能、位置以及它如何支持讨论部分将为你提供当前领域写作的参照。用一般现在时引用背景事实信息的句子数量和位置将使你很好地了解此类信息在讨论部分中的每个要点的常见程度和必要性。

句10："In addition the study provides evidence-based guidance for pragmatic public health recommendations that include age-related follow-up and treatment of post-V2D complications." **作者以研究的应用强有力地作结**。

4.2.3 讨论部分的模板

现在将下面的句子描述与你在第 4.2.1 节练习 1 中的列表进行比较：

句1，作者	明确指出研究对现有文献和知识的贡献
句2，作者	确定了该研究与已有文献相关的具体新特征，并总结了研究结果
句3 和句4，作者	回顾结果并做出评论
句5，作者	提及从结果中产生的潜在应用
句6、7 和8，作者	提及研究的局限性，并对未来的研究提出建议
句9，作者	在文献中提及背景事实信息以支持研究结论
句10，作者	以研究的应用强有力地作结

这些可以被简化以创建一个基本的、通用的讨论模板（见下表）。**将此模板与你在第4.2.1节中创建的组件列表合并，为你当前研究领域的讨论部分生成一组稳健的模板组件，你将随着研究的发展进行调整或更新。**组件的顺序相当流畅，然而，列表顶部的内容往往出现在讨论的前面，而底部的内容往往出现在稍后或接近尾声的位置。

通用讨论模板
宣布讨论部分的结构或内容
陈述研究的成就/贡献*

回顾背景信息/文献以“重启”读者
回顾空白/目标/方式
回顾结果并探索它们的含义
映射到文献/知识以进行比较/支持
指出潜在的局限性和对未来工作的建议
重申研究的成就/贡献/影响
确定潜在的应用

*注：研究的**成就/贡献**是最自由流动的组成部分——在某些情况下，它发生在开始和结束时，就像在这个模板中一样；在其他情况下，它发生在整个讨论中。

▶ 成就和贡献有区别吗?

有时候它们是一样的。成就是研究的内部成果，并和引言中所述的具体研究问题的成功解决是相关的。例如，如果研究旨在鉴定（identify）某事，那么鉴定（identifying）它就是研究的成就。作者使用相同的动词比切换到所谓的同义词，如查找（find）或检测（detect）更容易使读者跟踪。贡献更为外向，本质上是研究如何在应用或知识方面影响现实或研究世界。

4.3　验证和调整基本通用模板

4.3.1　模型的演示

在下面的讨论部分中识别模板组件。举例说明模板组件的

语言以**粗体**显示。注意模板是如何为讨论部分创建一个叙述框架的。

Rapid alloy prototyping: compositional and thermo-mechanical high throughput bulk combinatorial design of structural materials based on the example of 30Mn–1.2C–xAl triplex steels[p]

5. Discussion

5.1 Microstructure and mechanical properties of the 30Mn–1.2C–xAl steels

The **novel** bulk RAP approach **introduced in this work provides, for the first time**, a systematic evaluation of the compositional and thermomechanical trends associated with a change in the Al content of a group of Triplex steels with high Mn and C concentrations. **We observed that** without the addition of Al to the 30Mn–1.2C steels **the most favourable mechanical properties were obtained for** the as-homogenised state (Figs. 2 and 3a). **The observed properties are in reasonable agreement with data reported for** Mn–C alloyed TWIP steels of a **similar** chemical composition **[27, 28]. The observed embrittlement during ageing can be related to** the formation of the coarse, pearlitic particles on the grain boundaries **[27]** (Fig. 5a).

High amounts of Al (~8 wt.%), on the other hand, **result in** pronounced strengthening during ageing, depending on the time and temperature (Figs. 2 and 3c), and no coarse particles **could be observed in this case** (Fig. 5c). **In the light of previous results this typical** precipitation hardening behaviour, which **allows** tuning of the strength and ductility, **can be explained by** the formation and growth of j carbides during ageing **[17, 18, 21]. Due to their reportedly small** size, which is of the order of several nanometres, the j carbides **could not be reliably detected or identified** in the high throughput RAP OM observations conducted in this study. The darker particles visible in Fig. 5c **might be linked to** j carbides.

Alloys with intermediate Al concentrations (about 2–6 wt.%) **do not offer** mechanical properties on the same level compared with the aforementioned extreme cases under their respective optimal conditions (i.e. after the respective most suitable ageing treatments). On the other hand, a **much smaller** influence of the ageing parameters on tensile behaviour **can be observed in these cases** (Figs. 2 and 3b).

Within the limitations of this study (confined range of applied heat treatments, OM investigations, etc.) **this improved stability** of the mechanical properties during thermal exposure **can be attributed to** a concerted formation of j carbides and grain boundary pearlite, balancing the strengthening and embrittlement effects of intermediate amounts of Al.

In general it should be underlined that detailed investigations

of the role of j carbides and pearlite particles on the deformation mechanisms, as well as the precipitation type and the structural nature of the j carbides (i.e. spinodal vs. nucleation/growth), **require** higher resolution techniques, such as TEM or APT. **Nonetheless**, the mechanical data on both RAP (Figs. 2 and 3) and conventionally synthesised and processed alloys (Fig. 6a) **are in good agreement with previously reported values** for Fe–Mn–Al–C steels [17,18]. The RAP results **suggest that future efforts** regarding more detailed nanostructural investigations **should focus on** such high-C triplex steels with high Al concentrations (>8 wt.%), as they **offer the possibility of** covering the widest range of mechanical properties via ageing treatments (scalability, Fig. 2) and exhibit the lowest possible specific weight of all such steels.

4.3.2 练习2：识别模板组件

这是来自不同学科的研究文章的另外两个讨论部分。它们已被编辑以达到合适的长度，省略部分被标记为 [...]。在第一个模板中，模板的标识符在讨论部分的前半部分以**粗体**显示。在一些情况下，动词用**粗体**表示以引起你对动词时态功能的注意。在讨论的后半部分，在模板的标识符下画线，然后在第二个讨论的整个模板的标识符下画线。

注意句子开始的方式如何揭示包含信息的叙述骨架。

1 Sensory nerve induced inflammation contributes to heterotopic ossifification[r]

Discussion

Heterotopic ossifification **is** a disorder involving rapid bone formation within muscle, tendon, and ligaments, adjacent to skeletal bone, and **it has been linked to** an elevation in BMP2 signaling **[Shore and Kaplan, 2010]. Further**, the incidence of HO **appears to be** dramatically increased in individuals who have sustained traumatic injury to the nervous system **[Forsberg *et al.*, 2009]. Here we determined whether** localized changes in BMP signaling, which lead to heterotopic bone formation, can also alter peripheral nerve signaling through induction of neuroinflflammation. **Our results suggest that** in the presence of BMP2, sensory neurons **express** mediators of neuroinfflflammation, resulting in the recruitment of mast cells and remodeling of the nerve structure.

BMP2 **has been shown previously to** induce the expression of the neuroinflflammatory mediators, substance P and CGRP, in sensory neuron cultures **[Bucelli *et al.*, 2008]. Here we quantifified** changes in these mediators, in vivo, after delivery of cells expressing BMP2. **We found** a **signifificant and immediate** elevation of both proteins, in relation to the control, which **received** the same cells transduced with an Adempty virus. **Interestingly, we observed** a **strong**

correlation in elevation of these mediators, **immediately** following our delivery of BMP2. **However**, as the process continued over time, **we observed** a cyclical pattern in the expression of these mediators, with a **signifificant** decline in expression on day 2, followed by a **signifificant** rise in expression on day 3, and a trend towards another increase in expression by day 6. **Although** BMP2 would presumably be expressed for the fifirst 3–4 days, prior to the rapid clearance of the cells **[Fouletier-Dilling *et al.*, 2007]**, the kinetics of BMP2 receptor signaling in this model **is unclear. Intriguingly**, one of the fifirst steps **is** the rapid formation of brown adipocytes within the tissues **[Olmsted-Davis *et al.*, 2007]. We have previously shown** brown adipocytes to be necessary for patterning of the new bone, by their unique capacity to regulate the oxygen microenvironment, not only by stimulation of new vessels, but also by uncoupling of aerobic respiration and "burning" of oxygen **[Olmsted-Davis *et al.*, 2007]**. The result of this uncoupling **is** a release of energy as heat, which **could potentially** re-stimulate sensory neurons to respond and release substance P and CGRP. **This could potentially explain** the observed cyclical nature of the response, **suggesting** secondary or tertiary signaling events.

Performing the assay in animals lacking TRPV1, **we saw** a significant decrease in the volume of heterotopic bone formed, compared to animals with functional TRPV1. The suppression, rather than complete ablation, **suggests that** other TRPV family members present on sensory neurons

may also contribute to the induction of HO. **Although we do not rule out** alterations in other peripheral nerve signaling to the central nervous system in these animals, both substance P and CGRP **were found to be significantly** decreased in the TRPV1 mice. **We still observed a trend towards** an increase in substance P and CGRP upon addition of the AdBMP2 transduced cells. **However, this was not** above the normal background levels observed in wild type mice, nor was it statistically significant for CGRP, and the induction was over three folds lower for SP, **so it is unclear whether this** contributes to HO. **The result that** BMP2 **does** not induce SP or CGRP in TRPV1 mice **is not surprising, as it has been previously shown that** TRPV1 **induces** SP in response to capsaicin **[Theriault *et al.*, 1979]** and that TRPV1 also **controls** heat- and acid-induced CGRP release from sensory nerves **[Kichko and Reeh, 2009]. In addition, previous studies have revealed** decreased injury-induced neuropeptide release in TRPV1 mice **[Wang and Wang, 2005]**. While TRPV1 **is unquestionably involved** in pain and neuroinflammation, TRPV **has also been found** to be involved in diabetes **[Razavi *et al.*, 2006]** and obesity **[Motter and Ahern, 2008]**. Whether this is by the same mechanism proposed here, or by alternative mechanisms, **remains undetermined. Consequently**, deletion of TRPV1 **could have** additional pleiotropic effects. TRPV1 mice receiving Adempty transduced cells **did not produce** heterotopic bone, **which is in line with our previous findings** that Adempty

transduced cells **have not produced** HO in any animal model we have tested **[Olmsted-Davis *et al.*, 2002]**.

Mast cells **are known to be** recruited to nerves during times of neuroinflammation. Upon degranulation, mast cells **release** a number of digestive factors, chymases, tryptases, and other enzymes, which can cleave proproteins, leading to their activation. **These factors appear to be essential** for tissue remodeling of not only the nerve, but also other surrounding tissues, including the vasculature **[Johnson *et al.*, 1988; Richardson and Vasko, 2002; Kleij and Bienenstock, 2005; Schaible *et al.*, 2005; Kulka *et al.*, 2008].** Nerve remodeling **is thought to be** part of neurite outgrowth, or the ability to remodel and extend neurons. **Perhaps** this process is utilized to innervate the newly forming HO. **Alternatively, Adameyko *et al.* [2009] recently demonstrated** the presence of a stem cell population residing within peripheral nerves that would migrate from the nerve to undergo melanocyte differentiation. **We quantified** the number of mast cells after induction of HO **and found a significant** elevation in this population within 48 h, when compared to tissues receiving the control cells. **We observed** an upward trend in the number of mast cells on all days. **However, perhaps due to** the immune response evoked to clear the injected cells, **there was** also an increase in mast cell numbers in the control tissues, leading to a signifificant difference only on day 2. **Further, we observed** the mast cells, within the fifirst 48 h, associating with the nerves and within the nerves, as compared to control

tissues where the mast cells were usually located randomly throughout the tissues. **It is intriguing that we observed** the most signifificant difference at these early stages, since **this appears to parallel our findings** for the release of SP and CGRP within the tissues, **suggesting** mast cells **may be** recruited after release of these factors.

[...]

We next looked at whether the nerve remodeling was releasing cells that were essential to bone formation. As noted above, it has been previously demonstrated that precursors in peripheral nerves are the origin of skin melanoctyes. Therefore, mast cell degranulation, and subsequent nerve remodeling, was blocked using cromolyn, and we observed a significant decrease in HO. We next analyzed the nerves from these animals and found an increase in cells expressing markers of early stem cells (nanog and Klf4). These primitive markers were sporadic in the nerves of untreated animals, but completely covered the nerve in the cromolyn treated animals. This not only suggests that the early tissue changes lead to expansion of these cells, but also that the pool size of these cells within the nerve of untreated animals is extremely low, due to concomitant and rapid migration and differentiation. However, blockade of these latter steps with cromolyn leads to accumulation of these cells expressing primitive markers within the nerve. We note here, however, that the mechanisms of cromolyn action are incompletely understood. Although cromolyn is widely characterized as a "mast cell-stabilizer"

(i.e., an agent that blocks the release of mast cell mediators following appropriate activation of the cell) that can suppress mouse mast cell function in vivo, its molecular targets are neither fully defined nor restricted to mast cells [Galli *et al.*, 2008]. Moreover, while the mechanism of action of cromolyn mainly involves mast cell degranulation [Cox, 1967], other mechanisms, such as inhibition of neutrophils and eosinophil induced chemotaxis [Bruijnzeel *et al.*, 1990], have also been described.

To our surprise, we observed osterix positive cells on the nerve as early as day 2, in the presence of cromolyn. There were also cells that expressed primitive stem cell factors, which appeared to simultaneously express osterix, suggesting that these cells are osteoblast precursors. The majority of osterix positive expression was associated with the nerve. We also observed Klf-4^{+} and $nanog^{+}$ cells that were not associated with osterix, suggesting that these cells may have other potentials. Besides osteoblasts, another possible fate of these cells may be brown adipocytes, which we have shown previously to be critical for reduction of the oxygen tension in the microenvironment for cartilage formation [Olmsted-Davis *et al.*, 2007] and for secreting VEGF for vessel formation [Dilling *et al.*, 2010]. It has recently been noted that the Misty mouse phenotype [Sviderskaya *et al.*, 1998], which is defificient in brown fat, is caused by a mutation in dock 7 [Blasius *et al.*, 2009], a neuronal factor that regulates Schwann cell migration and neuronal polarity. It is intriguing

to speculate that brown fat progenitors may also reside in peripheral nerves, particularly since TRPV1 responds to heat [Szallasi *et al.*, 2007]. Additionally, it is interesting that the mutation in a single neuronal protein, dock7, not only dramatically increases HO in the Misty mouse [Olmsted-Davis *et al.*, 2007], but also causes severe osteoporosis in the skeletal bone (Rosen C., unpublished). Further, we previously demonstrated the rapid formation of new vessels early after BMP2 induction [Dilling *et al.*, 2010], suggesting that several types of tissues are being assembled simultaneously during this period. Osterix has previously been suggested to play a role in osteoblast lineage commitment of progenitors, suppressing the adipose phenotype [Cheng *et al.*, 2003]. Perhaps the early osterix expression, 4 days prior to the appearance of osteoid matrix, may be part of a regulatory mechanism to preserve these cells for future osteogenic fate. Finally, although not highlighted in this manuscript, we did observe osterix positive cells, at later times, in vessel-like structures that co-aligned with early endothelial markers, such as flk 1, which we have previously identified as characteristic of this early vasculogenesis [Dilling *et al.*, 2010]. This notion supports the work of Lounev *et al.* [2009], suggesting that osteoblast progenitors reside within the newly forming vessels, have a Tie 2 marker, and are not derived from marrow [Kaplan *et al.*, 2007]. Other investigators have also provided evidence for this concept of osteoblast progenitors being associated with the vasculature [Kolf *et al.*, 2007; Medici *et al.*, 2010].

This study is the first step in identifying a potential direct role for the peripheral nervous system in the induction of heterotopic ossification. The data suggest that early neuroinflammation, elicited in the presence of BMP2, may be capable of expanding a population of cells within the nerve, which can migrate and potentially contribute to a number of structures, rapidly assembling to produce HO. Suppression of these steps significantly decreases HO formation. Although it is unclear what effects this may have on the adjacent skeletal bone, the data suggest that there is direct communication with the hypothalamus, which could, in part, signal to impact bone remodeling. Understanding these earliest steps of HO will, for the first time, provide us novel targets for therapeutic intervention, which may ultimately lead to effective treatments. Finally, it is conceivable that such a mechanism could play a role in many other disease states, including neurofibromatosis and vascular calcification.

2 Supervised learning in spiking neural networks with FORCE training[8]

Discussion

We have shown that FORCE training can take initially chaotic networks of spiking neurons and use them to mimic the natural tasks and functions demonstrated by populations of neurons. For example, these networks were trained to learn

low-dimensional dynamical systems, such as oscillators which are at the heart of generating both rhythmic and non rhythmic motion[45]. We found FORCE training to be robust to the spiking model employed, initial network states, and synaptic connection types.

Additionally, we showed that we could train spiking networks to display behaviors beyond low-dimensional dynamics by altering the supervisor used to train the network. For example, we trained a statistical classifiifier with a network of Izhikevich neurons that could discriminate its inputs. Extending the notion of an oscillator even further allowed us to store a complicated sequence in the form of the notes of a song, reproduce the singing behavior of songbirds, and encode and replay a movie scene. These tasks are aided by the inclusion of a high-dimensional temporal signal (HDTS) that discretizes time by segregating the neurons into assemblies.

FORCE training is reminiscent of how songbirds learn their stereotypical learned songs[35,46]. Juvenile songbirds are typically presented with a species specifific song or repertoire of songs from their parents or other members of their species. These birds internalize the original template song and subsequently use it as an error signal for their own vocalization[35–37,39,46–49]. Our model reproduced the singing behavior of songbirds with FORCE training as the error correction mechanism. Both the spiking statistics of area RA and the song spectrogram were accurately reproduced after FORCE training. Furthermore, we demonstrated that altering

the balance between excitation and inhibition post training degrades the singing behavior post-training. A shift to excess excitation alters the spectrogram in a highly non-linear way while a shift to excess inhibition reduces the amplitude of all frequencies.

Inspired by the clock-like input pattern that songbirds use for learning and replay[35,36] we used a similar HDTS to encode a longer and more complex sequence of notes in addition to a scene from a movie. We found that these signals made FORCE training faster and the subsequent replay more accurate. Furthermore, by manipulating the HDTS frequency we found that we could speed up or reverse movie replay in a robust fashion. We found that compressing replay resulted in higher frequency oscillations in the mean population activity. Attenuating the HDTS decreased replay performance while transitioning the mean activity from a 4–8Hz oscillation to a slower (≈2Hz) oscillation. Finally, replay of the movie was robust to lesioning neurons in the replay network.

While our episodic memory network was not associated with any particular hippocampal region, it is tempting to conjecture on how our results might be interpreted within the context of the hippocampal literature. In particular, we found that the HDTS conferred a slow oscillation in the mean population activity reminiscent of the slow theta oscillations observed in the hippocampus. The theta oscillation is strongly associated to memory; however, its computational role is not fully understood, with many theories proposed[50–53]. For

example, the theta oscillation has been proposed to serve as a clock for memory formation[50,54].

Here, we show a concrete example that natural stimuli that serve as proxies for memories can be bound to an underlying oscillation in a population of neurons. The oscillation forces the neurons to fifire in discrete temporal assemblies. The oscillation (via the HDTS) can be sped up, or even reversed resulting in an identical manipulation of the memory. Additionally, we found that reducing the HDTS input severely disrupted replay and the underlying mean population oscillation. This mirrors experimental results that showed that theta power was predictive of correct replay[55]. Furthermore, blocking the HDTS prevents learning and prevents accurate replay with networks trained with an HDTS present. Blocking the hippocampal theta oscillation pharmacologically[56] or optogenetically[57] has also been found to disrupt learning.

The role of the HDTS is reminiscent of the recent discovery of time cells, which also serve to partition themselves across a time interval in episodic memory tasks[58–60]. How time cells are formed is ongoing research however they are dependent on the medial septum, and thus the hippocampal theta oscillation[61]. Time cells have been found in CA1[58], CA3[62] and temporally selective cells occur in the entorhinal cortex[63].

In a broader context, FORCE trained networks could be used in the future to elucidate hippocampal functions. For example, future FORCE trained networks can make use

of biological constraints such as Dale's law in an effort to reproduce verified spike distributions for different neuron types with regards to the phase of the theta oscillation[64]. These networks can also be explicitly constructed to represent the different components of the well studied hippocampal circuit.

FORCE training is a powerful tool that allows one to use any sufficiently complicated dynamical system as a basis for universal computation. The primary difficulty in implementing the technique in spiking networks appears to be controlling the orders of magnitude between the chaos inducing weight matrix and the feedback weight matrix. If the chaotic weight matrix is too large in magnitude (via the G parameter), the chaos can no longer be controlled by the feedback weight matrix[1]. However, if the chaos inducing matrix is too weak, the chaotic system no longer functions as a suitable reservoir. To resolve this, we derived a scaling argument for how Q should scale with G for successful training based on network behaviors observed in ref. [1]. Interestingly, the balance between these fluctuations could be related to the fading memory property, a necessary criterion for the convergence of FORCE trained rate networks[65].

Furthermore, while we succeeded in implementing the technique in other neuron types, the Izhikevich model was the most accurate in terms of learning arbitrary tasks or dynamics. This is due to the presence of spike frequency adaptation variables that operate on a much slower time scale than the neuronal equations. There may be other biologically relevant

forces that can increase the capacity of the network to act as a reservoir through longer time scale dynamics, such as synaptic depression and NMDA mediated currents for example[66–68].

Furthermore, we found that the inclusion of a high-dimensional temporal signal increased the accuracy and capability of a spiking network to reproduce long signals. In ref. [2], another type of high-dimensional supervisor is used to train initially chaotic spiking networks. Here, the authors use a supervisor consisting of $O(N^2)$ components (see ref. [2] for more details). This is different from our approach involving the construction of an HDTS, which serves to partition the neurons into assemblies and is of lower dimensionality than $O(N^2)$. However, from ref. [2] and our work here, increasing the dimensionality of the supervisor does aid FORCE training accuracy and capability. Finally, it is possible that an HDTS would facilitate faster and more accurate learning in networks of rate equations and more general reservoir methods as well.

Although FORCE trained networks have dynamics that are starting to resemble those of populations of neurons, at present all top-down procedures used to construct any functional spiking neural network need further work to become biologically plausible learning rules[1,5,8]. For example, FORCE trained networks require non-local information in the form of the correlation matrix $\mathbf{P}(t)$. However, we should not dismiss the final weight matrices generated by these techniques as biologically implausible simply because the techniques are themselves biologically implausible.